

Mark Ghebrial

☎ (949) 378-9520 ✉ mark.ghebrial@gmail.com
🌐 [linkedin.com/MarkGhebrial](https://www.linkedin.com/MarkGhebrial) 🐙 github.com/MarkGhebrial 🌐 ghebrial.net

Education

UC Riverside

September 2022 - June 2026

Bachelor of Science in Computer Engineering

Riverside, CA

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Logic Design, Embedded Systems, Operating Systems, Software Validation and Testing, Computer Networks, Circuit Analysis, Databases, Computer Graphics.
- **Club Participation:** Cybersecurity club, Rocketry club.

Projects

Mailroom | *Rust, tokio, sqlite, sea-orm*

github.com/MarkGhebrial/mailroom

- A from-scratch email server, written in Rust. Built because I wanted to learn how email works.
- Leveraged tokio and Rust async/await to efficiently handle simultaneous incoming and outgoing connections.
- Implemented the POP3 protocol, closely referencing published RFCs to ensure maximum compatibility with email clients. Currently working on an SMTP implementation.
- Designed database schema for storing users and user credentials.

GitBuddy | *C++, Git, gtest*

github.com/MarkGhebrial/GitBuddy

- A command-line utility for simplifying the Git workflow.
- Employed Scrum methodologies to lead a team of 4 people, delegating and splitting work into 1 week sprints.
- Constructed UML diagrams, navigation diagrams, and technical specifications to guide project development.
- Wrote unit tests using GoogleTest, enabling automatic testing of program behavior.

Closed Loop Nerf Blaster Controller | *C++, KiCad, Fusion 360*

github.com/MarkGhebrial/blaster_control

- Custom electronics and code for a flywheel based 3D printed Nerf blaster.
- Designed custom PCB for IR reflectivity sensors. Wrote real-time embedded code targeted for Atmel SAMD chips that calculates wheel RPM based on the time between sensor pulses.
- Implemented a feedforward model to accurately represent motor behavior, enhancing PID controller effectiveness and enabling precise motor control from 1,000 to 40,000 RPM.

Opengyro | *Embedded Rust, KiCad*

github.com/MarkGhebrial/opengyro

- A fly-by-wire system for my radio controlled airplane.
- Designed a custom PCB featuring voltage regulator, servo control headers, and connector for I2C peripherals.
- Used Rust and a low level hardware abstraction layer to write firmware targeted for the Atmel SAMD51 Arm microcontroller, granting me fine grained control over CPU peripherals.
- Sped up UART reads by more than 100% by configuring Direct Memory Access.

Highlander Space Program Pad Controller | *KiCad*

github.com/Highlander-Space-Program/pad-controller-pcb

- A custom circuit board for controlling UC Riverside's liquid rocket engine test stand.
- Collaborated with teammates to determine board requirements and ensure smooth system integration.
- Designed a high power buck converter circuit, carefully spec'ing passive components to ensure correct operation.
- Assembled a prototype using a perfboard and careful soldering. Used a multimeter and oscilloscope for debugging and troubleshooting.

Technical Skills

Languages: Rust, C++, C, Python, Java, Arduino, Kotlin

Technologies: Git, GitHub, Linux, tokio, Make, Cmake

CAD Programs: KiCad, Onshape, Fusion 360, SolidWorks

Concepts: Object-oriented programming, Model-view-controller model, Operating systems, Memory mapped registers, Encryption & decryption, Assembly code, REST APIs, GPIO, PID Controllers, Computer networking

Experience

UC Riverside

January 2025 - March 2025

Grader, Advanced Software Testing and Analysis

Riverside, CA

- Graded assignments for UCR's grad level Software Testing and Analysis class, assessing 50+ students on their understanding of relevant research papers.

iPolymer

Summer of 2022

Administrative Assistant

Irvine, CA

- Performed various data entry, inventory management, and engineering tasks.
- Wrote a Python script to import data from a temperature probe into a .csv file.
- Designed and 3D printed a custom fixture for characterizing the thermal characteristics of a solenoid.

FIRST Robotics Competition - Team 3309

2018-2022

Director of Software Architecture/Lead Programmer

Anaheim, CA

- Led Team 3309's programming team for 3 years, using Git and GitHub to manage code versioning.
- Wrote sophisticated Java code for 4 robots and designed a Kotlin library to improve code ergonomics.
- Designed and implemented the control code for the team's custom omnidirectional swerve drivetrain, enabling fluid and precise robot movement.
- Served as robot technician in high pressure competition environments, completing pre and post match checklists to ensure maximum robot performance.